Assignment_01_CO_02

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QUESTIONS

- Q1 Check whether a number is an Emirp number — 5 marks
- Q2 Check whether a number is a Circular Prime — 5 marks
- Q3 Check whether a number is a Kaprekar number — 5 marks
- Q4 Sphenic Number — 5 marks
- Q5 Check whether a number is a Trimorphic number — 5 marks
- Q6 Check whether a number is a Keith number — 5 marks

6/6 answered

Q1 Check whether a number is an Emirp number

5 marks

Topics to use: Classes, objects, member functions, loops, digit reversal, prime checking, logical validation.

Question: An Emirp number is a prime number whose reversed number is a different prime number.

Example: 13
- 13 is prime.
- Reversed number = 31.
- 31 is also prime and differs from 13.
- Therefore, 13 is an Emirp number.

Test Cases:
1.13->Emirp number
2.23->Not an Emirp number
3.1->Invalid input

Your Code

```
#include <iostream>
using namespace std;

class Emirp
{
public:
    int prime(int n)
    {
        if (n < 2) return 0;
        for (int i = 2; i * i <= n; i++)
            if (n % i == 0) return 0;
        return 1;
    }

    void check(int n)
    {
        if (n < 2)
            cout << "Invalid input";
        else
        {
            int r = 0, t = n;
```

Input

```
13
```

Output

```
Emirp number
```

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

Keith number
5 marks

6/6 answered

```cpp
public:
    int prime(int n)
    {
        if (n < 2) return 0;
        for (int i = 2; i * i <= n; i++)
            if (n % i == 0) return 0;
        return 1;
    }

    void check(int n)
    {
        if (n < 2)
            cout << "Invalid input";
        else
        {
            int r = 0, t = n;
            while (t > 0)
            {
                r = r * 10 + t % 10;
                t /= 10;
            }

            if (prime(n) && r != n && prime(r))
                cout << "Emirp number";
            else
                cout << "Not an Emirp number";
        }
    }
};

int main()
{
    int n;
    cin >> n;
    Emirp obj;
    obj.check(n);
    return 0;
}
```

Submitted

Output

```
Emirp number
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

5.00 / 5 marks

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- Q6 — Check whether a number is a Keith number — 5 marks

6/6 answered

Q2 Check whether a number is a Circular Prime

5 marks

Topics to use: Classes, objects, member functions, loops, prime checking, digit rotation, arrays, logical validation.

Question: A Circular Prime is a prime number for which every number formed by rotating its digits is also prime.

Example: 197
- Original number: 197 → prime.
- Rotate left: 971 → prime.
- Rotate left again: 719 → prime.
- Therefore, 197 is a Circular Prime.

Test Cases:
1.197->Circular Prime
2.101->Not a Circular Prime
3.2->Circular Prime

Your Code

```cpp
#include <iostream>
using namespace std;

class Circular
{
public:
    int prime(int n)
    {
        if (n < 2) return 0;
        for (int i = 2; i * i <= n; i++)
            if (n % i == 0) return 0;
        return 1;
    }

    void check(int n)
    {
        int t = n, p = 1, count = 0;

        while (t > 0)
        {
```

Input

```
197
```

Output

```
Circular Prime
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

6/6 answered

```cpp
        if (n < 2) return 0;
        for (int i = 2; i * i <= n; i++)
            if (n % i == 0) return 0;
        return 1;
    }

    void check(int n)
    {
        int t = n, p = 1, count = 0;

        while (t > 0)
        {
            count++;
            p *= 10;
            t /= 10;
        }
        p /= 10;

        for (int i = 0; i < count; i++)
        {
            if (!prime(n))
            {
                cout << "Not a Circular Prime";
                return;
            }
            n = (n % p) * 10 + n / p;
        }

        cout << "Circular Prime";
    }
};

int main()
{
    int n;
    cin >> n;

    Circular obj;
    obj.check(n);

    return 0;
}
```

Submitted

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

5.00 / 5 marks

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- Q6 Check whether a number is a Keith number — 5 marks

6/6 answered

Q3 Check whether a number is a Kaprekar number

5 marks

Topics to use: Classes, objects, member functions, loops, digit counting, arithmetic operations, conditional logic.

Question: A positive number is a Kaprekar number if its square can be split into two parts such that the sum of those parts equals the original number. The right part must contain as many digits as the original number.

Example: 45
- Square = \(45^2 = 2025\)
- Split into 20 and 25.
- Sum = \(20 + 25 = 45\).

Test Cases:
1.9->Kaprekar number
2.45->Kaprekar number
3.10->Not a Kaprekar number

Your Code

```cpp
#include <iostream>
using namespace std;

class Kaprekar
{
public:
    void check(int n)
    {
        if (n <= 0)
        {
            cout << "Invalid input";
            return;
        }

        int sq = n * n, t = n, p = 1;

        while (t > 0)
        {
            p *= 10;
            t /= 10;
```

Input

```
10
```

Output

```
Not a Kaprekar number
```

Visible Test Cases

- Test Case 1
- Test Case 2
- Test Case 3

Q6
Check whether a number is a Keith number
5 marks

6/6 answered

```cpp
{
public:
    void check(int n)
    {
        if (n <= 0)
        {
            cout << "Invalid input";
            return;
        }

        int sq = n * n, t = n, p = 1;

        while (t > 0)
        {
            p *= 10;
            t /= 10;
        }

        int right = sq % p;
        int left = sq / p;

        if (left + right == n)
            cout << "Kaprekar number";
        else
            cout << "Not a Kaprekar number";
    }
};

int main()
{
    int n;
    cin >> n;

    Kaprekar obj;
    obj.check(n);

    return 0;
}
```

Submitted

Output

```
Not a Kaprekar number
```

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

5.00 / 5 marks

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QUESTIONS

- Q1 ✓ Check whether a number is an Emirp number — 5 marks
- Q2 ✓ Check whether a number is a Circular Prime — 5 marks
- Q3 ✓ Check whether a number is a Kaprekar number — 5 marks
- Q4 ✓ Sphenic Number — 5 marks
- Q5 ✓ Check whether a number is a Trimorphic number — 5 marks
- Q6 ✓ Check whether a number is a Keith number — 5 marks

6/6 answered

Q4 Sphenic Number

5 marks

Topics: Classes and objects, access specifiers, constructors, member functions, loops, prime factorization, and arrays of objects.
Write a C++ program using a class named NumberAnalyzer to check whether a positive integer is a Sphenic Number. A Sphenic Number is a number that can be expressed as the product of exactly three distinct prime numbers, each appearing only once.

The program must identify the prime factors, count them, and verify that exactly three distinct prime factors multiply to the original number.

Example:

Input: 30
\[
30=2\times3\times5
\]

Since 2, 3, and 5 are three distinct primes, 30 is a Sphenic Number.

Test Cases:
1.30->Sphenic
2.42->Sphenic
3.210->Not Sphenic

Your Code

```cpp
#include <iostream>
using namespace std;

class NumberAnalyzer
{
    int n;

public:
    NumberAnalyzer(int x)
    {
        n = x;
    }

    void check()
    {
        int count = 0, t = n;
```

Input

```
30
```

Output

```
Sphenic
```

Visible Test Cases

Test Case 1 ✓

```cpp
    NumberAnalyzer(int x)
    {
        n = x;
    }

    void check()
    {
        int count = 0, t = n;

        for (int i = 2; i <= t; i++)
        {
            if (t % i == 0)
            {
                count++;
                t /= i;
                if (t % i == 0)
                    break;
            }
        }

        if (count == 3 && t == 1)
            cout << "Sphenic";
        else
            cout << "Not Sphenic";
    }
};

int main()
{
    int n;
    cin >> n;

    if (n <= 0)
        cout << "Invalid input";
    else
    {
        NumberAnalyzer obj(n);
        obj.check();
    }

    return 0;
}
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

5.00 / 5 marks

Submitted

QUESTIONS

Q1
Check whether a number is an Emirp number
5 marks

Q2
Check whether a number is a Circular Prime
5 marks

Q3
Check whether a number is a Kaprekar number
5 marks

Q4
Sphenic Number
5 marks

Q5
Check whether a number is a Trimorphic number
5 marks

Q6
Check whether a number is a Keith number
5 marks

6/6 answered

Q5 Check whether a number is a Trimorphic number

5 marks

Topics to use: Classes, objects, private data members, member functions, loops, arithmetic operations, remainder operator.

Question: A number is called a Trimorphic number if its cube ends with the original number.

Example: 24

- Cube = \(24^3 = 13824\)
- The last two digits are 24.
- Therefore, 24 is a Trimorphic number.

Test Cases:

1.24->Trimorphic number

2.5->Trimorphic number

3.7->Not a Trimorphic number

Your Code

```cpp
#include <iostream>
using namespace std;

class Trimorphic
{
    int n;

public:
    void check(int x)
    {
        n = x;
        int cube = n * n * n, p = 1, t = n;

        while (t > 0)
        {
            p *= 10;
            t /= 10;
        }

        if (cube % p == n)
            cout << "Trimorphic number";
        else
            cout << "Not a Trimorphic number";
    }
};
```

Input

```
24
```

Output

```
Trimorphic number
```

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

5.00 / 5 marks

5 marks

Q6
Check whether a number is a Keith number
5 marks

6/6 answered

```cpp
class Trimorphic
{
    int n;

public:
    void check(int x)
    {
        n = x;
        int cube = n * n * n, p = 1, t = n;

        while (t > 0)
        {
            p *= 10;
            t /= 10;
        }

        if (cube % p == n)
            cout << "Trimorphic number";
        else
            cout << "Not a Trimorphic number";
    }
};

int main()
{
    int n;
    cin >> n;

    if (n < 0)
        cout << "Invalid input";
    else
    {
        Trimorphic obj;
        obj.check(n);
    }

    return 0;
}
```

Submitted

Output

```
Trimorphic number
```

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

5.00 / 5 marks

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- **Q4** Sphenic Number — 5 marks
- **Q5** Check whether a number is a Trimorphic number — 5 marks
- **Q6** Check whether a number is a Keith number — 5 marks

6/6 answered

Q6 Check whether a number is a Keith number

5 marks

Topics to use: Classes, objects, member functions, loops, arrays, digit extraction, sequence generation.

Question: A Keith number is a positive integer whose digits form the beginning of a sequence. Each subsequent term is the sum of the previous terms, using as many terms as the original number has digits. If the sequence reaches the original number, it is a Keith number.

Example: 197

Starting digits: 1, 9, 7
- \(1+9+7=17\)
- \(9+7+17=33\)
- \(7+17+33=57\)
- \(17+33+57=107\)
- \(33+57+107=197\)

Therefore, 197 is a Keith number.

Test Cases:
1.197->Keith number
2.742->Keith number
3.7->Not a Keith number

Your Code

```cpp
#include <iostream>
using namespace std;

class Keith
{
public:
    void check(int n)
    {
        if (n < 10)
        {
            cout << "Not a Keith number";
            return;
        }

        int a[10], t = n, c = 0, s;
```

Input

```
7
```

Output

```
Not a Keith number
```

Visible Test Cases

```
        int a[10], t = n, c = 0, s;

        while (t > 0)
        {
            a[c++] = t % 10;
            t /= 10;
        }

        for (int i = 0; i < c / 2; i++)
            swap(a[i], a[c-i-1]);

        while (true)
        {
            s = 0;
            for (int i = 0; i < c; i++)
                s += a[i];

            if (s == n)
            {
                cout << "Keith number";
                return;
            }

            if (s > n) break;

            for (int i = 0; i < c-1; i++)
                a[i] = a[i+1];

            a[c-1] = s;
        }

        cout << "Not a Keith number";
    }
};

int main()
{
    int n;
    cin >> n;

    Keith obj;
    obj.check(n);

    return 0;
}
```

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

5.00 / 5 marks